

ENVS 193DS Homework 03

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```
# reading in packages
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.0      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.2      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr      1.2.1
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(here)
```

here() starts at /Users/luciamurillo/github/ENVS-193DS_homework-03

```
library(janitor)
```

Attaching package: 'janitor'

The following objects are masked from 'package:stats':

chisq.test, fisher.test

```
library(readxl)
# creating new object from kelp data
kelp <- read_csv((here::here ("data","temp-kelp.csv")))
```

Rows: 32 Columns: 2

-- Column specification -----

Delimiter: ","

dbl (2): temp_c, kelp_elong

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
# creating new object from personal data
personal_data <- read_csv((here::here ("data","personal-data.csv"))) |>
# cleaning names of columns
clean_names() |>
# dropping N/A values
drop_na(date)
```

New names:

Rows: 34 Columns: 10

-- Column specification

----- Delimiter: "," chr

(3): DATE, LOCATION, MUSIC (GENRE) dbl (4): TIME (#HOURS), # OF FRIENDS,
CAFFEINE LEVEL (1-5), PRODUCTIVITY RA... lgl (2): ...1, ...10 time (1): TIME OF
START

i Use `spec()` to retrieve the full column specification for this data. i

Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

* `` -> `...1`

* `` -> `...10`

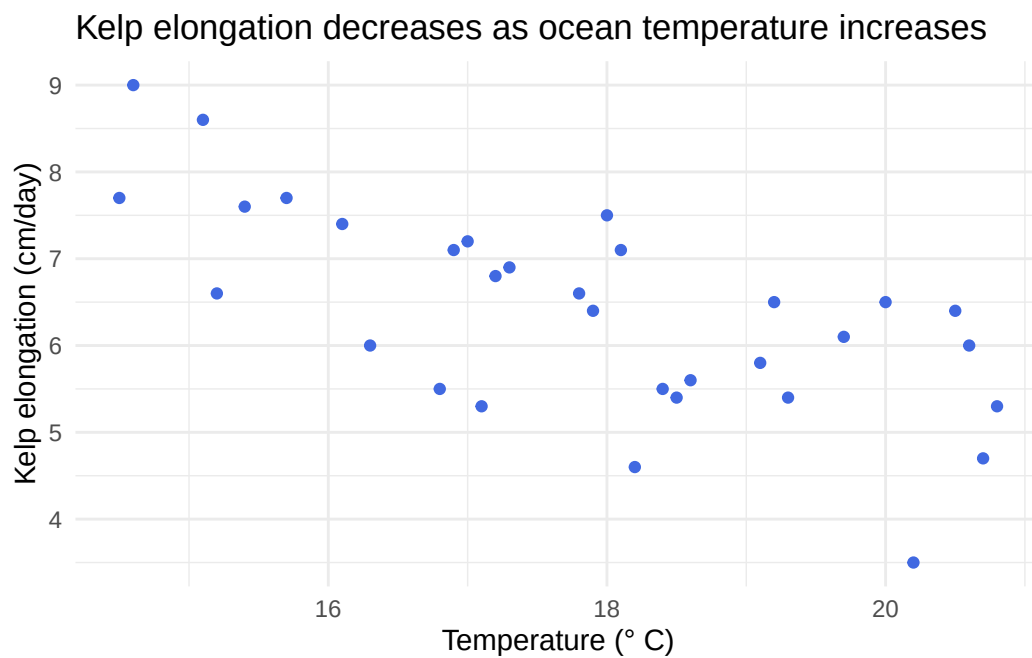
Problem 1: Giant Kelp fronds

(a)

One method of determining the strength of a relationship between temperature and giant kelp frond elongation is by finding the Pearson's correlation (r), which determines the direction and strength of the linear relationship between variables. The non parametric version is the Spearman rank correlation, which determines the monotonic relationship between variables.

(b)

```
# base layer ggplot
# data: kelp
ggplot(data = kelp,
       mapping = aes(
         # x-axis: temperature
         x = temp_c,
         # y-axis: kelp elongation
         y = kelp_elong)) +
# first layer: scatterplot
# changing color to royal blue
geom_point(col = "royalblue") +
# changing x and y axis titles, adding title
labs(x = "Temperature (\u00B0 C)",
     y = "Kelp elongation (cm/day)",
     title = "Kelp elongation decreases as ocean temperature increases") +
# changing to minimal theme
theme_minimal()
```



Do I need to check residuals?

(c)

Checking assumptions:

In part 1b, I created a scatterplot to observe the patterns displayed in the `kelp` data. In this plot, a negative linear relationship between ocean temperature and kelp elongation is visible. This indicates that finding the Pearson's correlation would be the best way to determine the strength of this relationship.

Running a test

```
# correlation test
# variables: temperature + kelp elongation
cor.test(~ temp_c + kelp_elong,
         # data from kelp dataframe
         data = kelp,
         # pearson correlation
         method = "pearson")
```

Pearson's product-moment correlation

```
data: temp_c and kelp_elong
t = -5.189, df = 30, p-value = 1.366e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.8359665 -0.4460157
sample estimates:
cor
-0.6877489
```

Make sure outputs are visible

(d)

To evaluate the strength of the relationship between ocean water temperature and kelp elongation, I used a Pearson's correlation test because the data is linear. I found a moderate negative (Pearson's $r = -0.69$, $t(30) = -5.19$, $p = 1.366e-05$, $0.05 < p < 0.01$) relationship between ocean temperature and kelp elongation.

(e)

I found that kelp elongation rates tend to decrease as temperatures increase. There is a moderate negative relationship between these two variables (Pearson's $r = -0.69$, $p = 1.366e-05$, $03B1 = 0.05$). **FINISH LATER ASK ABT DOING A LM**

(f)

```
# correlation test
# variables: temperature + kelp elongation
cor.test(~ temp_c + kelp_elong,
         # data from kelp dataframe
         data = kelp,
         # spearman correlation
         method = "spearman")
```

```
Warning in cor.test.default(x = mf[[1L]], y = mf[[2L]], ...): Cannot compute
exact p-value with ties
```

Spearman's rank correlation rho

```
data: temp_c and kelp_elong
S = 9216.1, p-value = 1.29e-05
alternative hypothesis: true rho is not equal to 0
sample estimates:
rho
-0.6891684
```

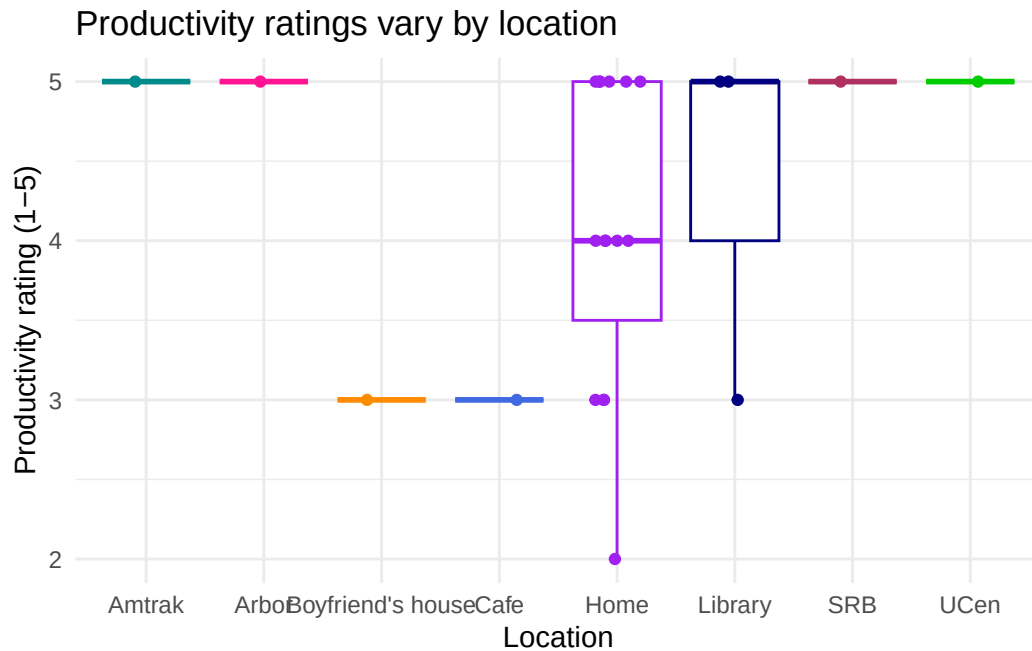
make sure results are displayed

These two tests have led me to the same interpretation of results. In the Spearman correlation test, I found a moderate negative relationship (Spearman's $U+03C1 = 0.69$, $S = 9216.1$, $p = 1.29e-05$, $03B1 = 0.05$) between temperature and kelp elongation.

Problem 2: Personal Data

(a)

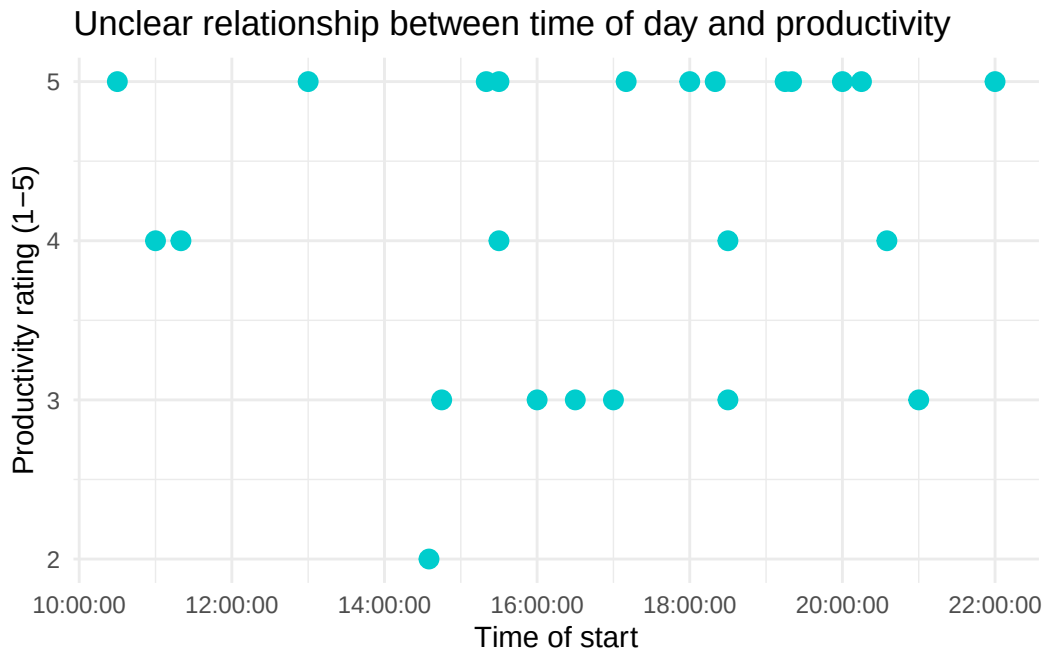
```
# ggplot base layer, data from personal_data
ggplot(data = personal_data,
      # x-axis: location
      # y-axis: productivity rating
      mapping = aes(x = location,
                    y = productivity_rating_1_5,
                    # color by location
                    color = location)) +
# boxplot layer 2
geom_boxplot() +
# jitter plot layer 3
geom_jitter(height = 0, width = 0.2) +
# changing titles of axes, adding title
labs(x = "Location",
      y = "Productivity rating (1-5)",
      title = "Productivity ratings vary by location") +
# adding minimal theme
theme_minimal() +
# changing colors of graph
scale_color_manual(values = c(
  "darkcyan", "deeppink", "darkorange", "royalblue", "purple", "navy", "maroon", "green3"))
# removing legend
theme(legend.position = "none")
```



add subtitle, wrap x-axis labels?, take out grids and background

```
# ggplot base layer, data is personal_data
ggplot(data = personal_data,
  # x-axis: time of start
  # y-axis: productivity rating
  mapping = aes(x = time_of_start,
    y = productivity_rating_1_5)) +

# add points
geom_point(
  # change color and size of points
  size = 3, color = "cyan3") +
#ADD geom_line(col = "cyan3") +??
# adding labels to axes and adding title
labs(x = "Time of start",
  y = "Productivity rating (1-5)",
  title = "Unclear relationship between time of day and productivity") +
# apply minimal theme
theme_minimal() +
# remove legend
theme(legend.position = "none")
```



add subtitle, change plot??, take out grid and background

(b)

Figure 1: Comparison of productivity levels at different locations. Each boxplot represents a location where I have studied, and each point represents a single study session.

Figure 2: The relationship between studying start time and productivity. Each point represents a single study session. **MAKE THAT ONE A LM??**

Problem 3: Affective visualization

Problem 4: Statistical critique

(a)

The test I have selected is the Mann-Whitney U-test. The response variable is treefrog density and the predictor variables are urbanization and crayfish presence. There is not a figure or table that is associated with the Mann-Whitney test that I am focusing on.

(b)

The test I am focusing on does not have a figure or a table.

```
knitr::include_graphics((here::here ("data","stream-screenshot.png")))
```

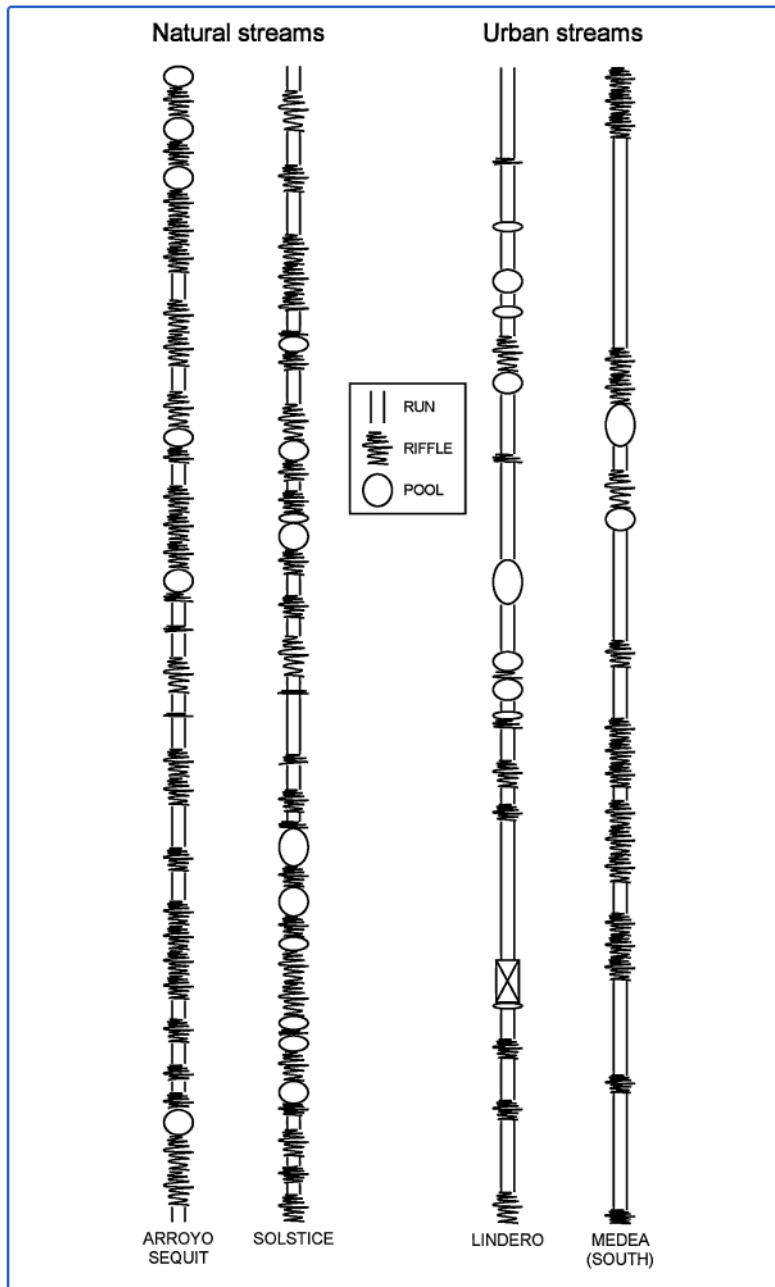


Figure 3

[Open in figure viewer](#) | [PowerPoint](#)

Schematic representation of habitat diversity (runs, riffles, and pools) in two urban and two natural streams in the Santa Monica Mountains and Simi Hills of southern California. The rectangle with an X on Lindero Creek represents a culvert.